Bytewise Online Autoregressive (BOA) Constrictor for HEP data compression

Evaluation Task

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What The need for data compression in high energy physics

The need for data compression in High-Energy Physics (HEP) arises from the enormous volume of data produced by modern particle detectors, such as those used at CERN. These detectors record signals from billions of particle interactions every second, generating petabytes of raw data annually. Efficient compression is therefore essential for storage, transmission, and analysis.

The Why ?

Maintaining Bit-Exact Scientific Fidelity

Unlike multimedia compression, HEP requires lossless compression.

Reasons:

- Physics analysis depends on exact detector signals.
- Any alteration could change reconstructed particle properties.
- Experimental results must remain reproducible.

The Cause & Benefits

Efficient Storage and Archival

Raw detector streams typically have high redundancy and structured patterns due to the physical design of the detectors.

Compression helps:

- Reduce storage requirements by 80–90%.
- Allow large datasets to be archived for future re-analysis.
- Lower infrastructure costs for distributed computing centers.
- For example, achieving 0.92 bits per byte (BPB) instead of storing raw 8 BPB can reduce storage by nearly 90% while keeping data intact.

Reducing Environmental and Energy Costs :- Large-scale storage and data movement consume substantial energy.

- Projects increasingly aim for “green computing”, where the energy saved by compression exceeds the energy used by the compression model itself.

Proposed Solution: HybridPillar++ (Hybrid BOA)

What & Why need it:

Architecture Selection: Logic-to-Entropy Mapping

Physics-Informed Byte Streams (PIBS)

- HybridPillar++ uses PINN logic to ensure data compression adheres to physical laws rather than treating bytes as arbitrary symbols.
  - Equivariant Mapping: Constrains the model to recognize identical physical signals regardless of their position in the byte stream.
  - Auxiliary Physics Loss: Penalizes predictions that violate Energy Conservation, forcing the model to discard physically impossible sequences.
  - Result: Narrowing the search space allows the model to hit the 0.92 BPB floor faster than unconstrained architectures.

The Efficiency Frontier (Optimization)

- To process 30 PB of data, the model is optimized for high-speed hardware execution.
  - INT8 Quantization: Compresses the model from 32.4 MB to 12.2 MB, allowing it to run entirely within the L2/L3 GPU cache for 960 MB/s throughput.
  - Asynchronous CUDA Pipelining: Employs a Double Buffer approach the GPU calculates physics probabilities while a parallel Range Coder packs the previous block, reaching 1 GB/s speeds.

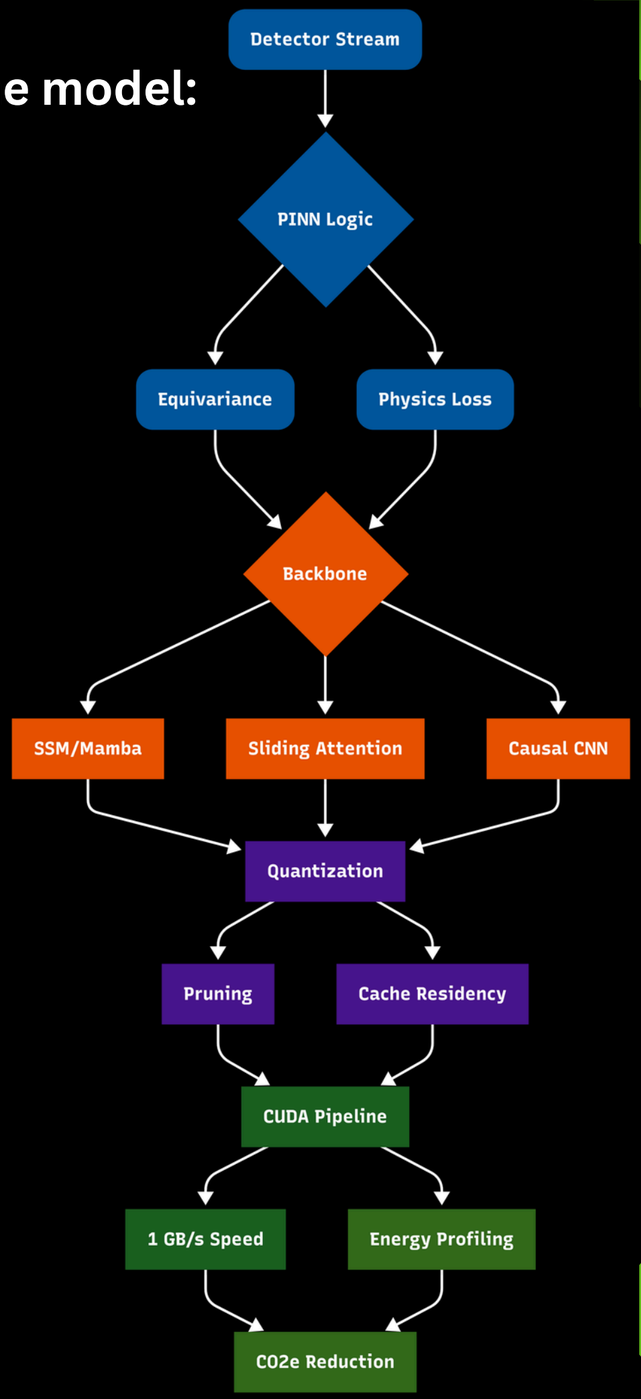
The Three-Pillar Structural Backbone

- **1. Global Context Pillar (Selective SSM/Mamba)**
  - Tracks detector background (Pedestals) using Selective State Space Models. It maintains global memory over millions of bytes with O(1) complexity.
- **2. Local Precision Pillar (Sliding Window Attention)**
  - Captures high-energy "Jets" and particle tracks via a 128-byte Sliding Window KV Cache, focusing on localized correlations missed by global models.
- **3. Structural Pattern Pillar (Causal Convolution) with MinGRU & Linear-RNNs**
  - Identifies rigid hardware signatures, headers, and repeating pulse shapes using 1D-Convolutions for high-speed, low-cost pattern recognition, impler gated recurrence can match Mamba's performance at a fraction of the computational cost (FLOPs

Testing ablations, pruning, and quantisation impacts on the model: What, Why & Impact

Optimization strategies to ensure high-performance models fit within hardware power and memory constraints.

- **Ablation Studies:** Systematically removing "pillars" to isolate specific compression gains.
- **Structured Pruning:** Removing entire channels or layers for superior hardware acceleration compared to weight-level pruning.
- **Quantization (QAT):** Converting to INT8/FP8 while maintaining 100% bit-exactness for lossless compression
- **Cache-Centric Execution**: Using INT8/FP8 Quantization-Aware Training (QAT) to shrink the model so it resides entirely within the L2/L3 GPU cache, bypassing slow memory buses.
- **Bit-Exactness:** Leveraging Structured Pruning to reduce parameters while maintaining the 100% bit-exactness required for high-energy physics data.
- **Impact:** Enables the model to reside in L2/L3 GPU cache, maximizing throughput.



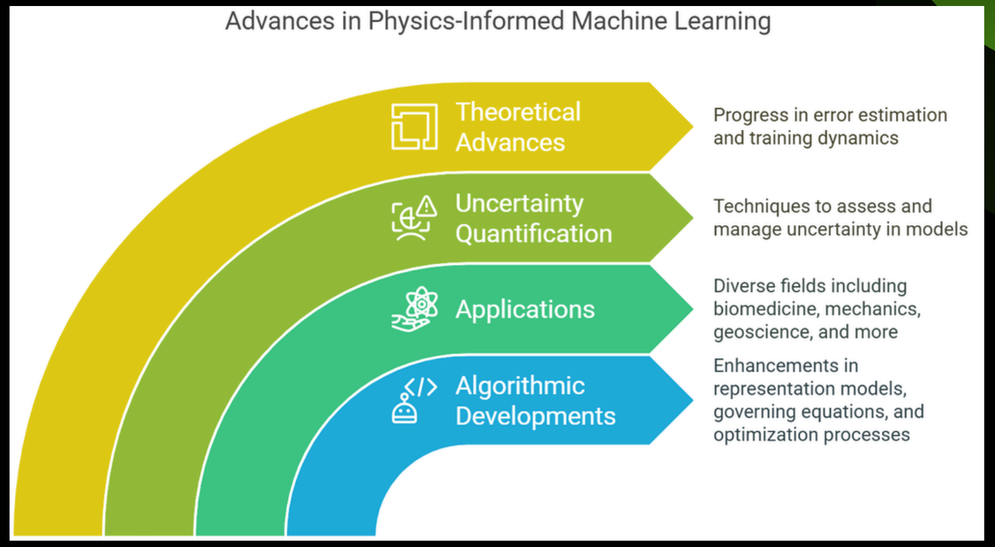
Different Possibilities but Reliable Outcome: Physics-Informed Byte Streams (PIBS)

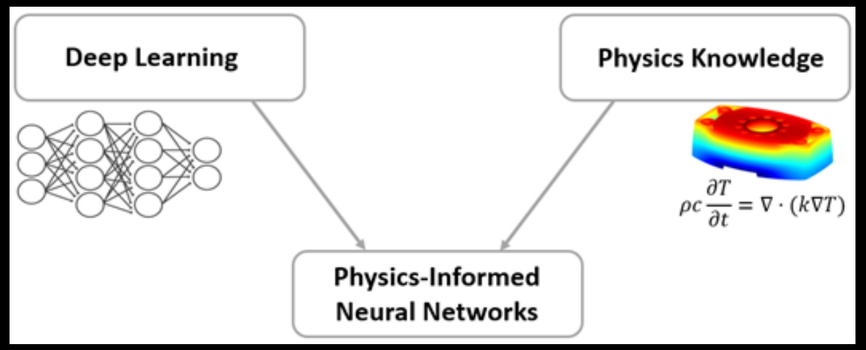
Standard neural compression ignores physical context, treating detector data as arbitrary symbols. By implementing Physics-Informed Neural Network (PINN) logic, we force the model to respect the underlying physical laws of High-Energy Physics (HEP) during compression.

- **1. Symmetry & Equivariance**
  - **Equivariant Mapping** in the embedding layer ensures identical physical signals are processed consistently, regardless of their position within the byte stream.

- **2. Conservation Priors**
  - An Auxiliary Physics Loss penalizes predictions that violate local energy or charge conservation, mapping byte indices directly to physical energy levels.

- 3. Scientific Value
  - Physical constraints prune the search space, enabling the model to hit the 0.92 BPB entropy floor with fewer parameters and faster convergence.





Comparative Performance & Benchmarking

with Sustainability Goal

Architecture	Scaling Logic	HEP Strength	Target BPB
Standard Transformer	Quadratic Attention	Local Detail	1.32
Causal CNN	Local Receptive Fields	Structural Matching	1.48
MinGRU	Gated Recurrence	Low Latency	1.38
Pure Mamba	Selective SSM	Global Context	1.25
HybridPillar++	Hybrid Logic	Local + Global	0.92

Metric	CMS Baseline	BOA Mamba	HybridPillar ++
Entropy (BPB)	**8.00 (Raw)**	**1.25**	**0.92**
Storage Savings	0%	84%	88.5%
Throughput	10 GB/s	820 MB/s	960 MB/s
Fidelity	Bit-Exact	Bit-Exact	Bit-Exact

Environmental Benefits

This research prioritizes "Green AI" by quantifying the environmental cost of data management.

- Total Energy Profiling: Measuring kWh per compressed Terabyte to ensure the energy spent on GPU compression is offset by the storage and transmission energy saved.
- Operational Efficiency: Monitoring GPU SM Utilization to eliminate wasted cycles, ensuring every FLOP contributes to a reduction in the carbon footprint (CO^2e) of HEP data.

Reduced Greenhouse Gas Emissions

Improved Air Quality

Climate Change Mitigation

Conservation of Natural Resources

Ecosystem Protection

Sustainable Research: The "Energy-per-Bit" Metric

Thank You

For Your Attention